

Md. Saifur Rahman

Software Engineer | AI Engineer

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PROFESSIONAL EXPERIENCE

Betopia Group

Dhaka, Bangladesh

Jr. AI Developer (Scale Up Ads Agency)

July 2025 – March 2026

- Developed scalable **ML-driven backend services** and **REST APIs** in Python to automate marketing workflows.
- Built **data pipelines** for large-scale advertising datasets, including feature extraction and preprocessing.
- Optimized **model scalability and inference performance** for production ML systems.

AI & MACHINE LEARNING PROJECTS

Bangla Sign Language Recognition System | *Python, Deep Learning, Computer Vision, Flutter*

- Developed a **real-time video-based Bangla Sign Language (BdSL)** recognition system using **CNN-based deep learning models**.
- Designed a complete **video processing pipeline** including frame extraction, hand detection, preprocessing, and gesture classification.
- Applied **relative quantization of hand landmarks** to normalize spatial relationships between keypoints, improving model robustness and gesture representation.
- Achieved efficient inference performance suitable for real-time applications.

Guava Disease Recognition System | *Python, TensorFlow, FastAPI, Flutter*

- Architected an end-to-end **computer vision pipeline** using **Convolutional Neural Networks (CNNs)** for automated fruit and leaf disease classification.
- Implemented dataset preprocessing, augmentation, and model training for improved classification accuracy.
- Deployed the model through a **FastAPI-based inference server** integrated with a Flutter mobile application.

IMDB Movie Review Sentiment Analysis | *Python, NLP, Scikit-learn, PyTorch/Keras*

- Developed and benchmarked multiple **NLP models** including **Logistic Regression, Naive Bayes, and LSTM networks**.
- Applied advanced preprocessing techniques such as **tokenization, TF-IDF vectorization, and word embeddings**.

Bangla Alphabet Handwritten Recognition | *Machine Learning, Python, Scikit-learn*

- Built a handwriting recognition system for Bangla characters using **supervised learning algorithms**.
- Performed image preprocessing, feature extraction, and model optimization to improve classification performance.

Real-Time Face Recognition System | *Python, OpenCV*

- Implemented a real-time face detection and recognition system using **Haar Cascade classifiers** and OpenCV.

TECHNICAL SKILLS

- **Programming Languages:** Python, C, C++, Java, Dart, SQL
- **AI & Machine Learning:** Machine Learning, Deep Learning, Natural Language Processing (NLP), Computer Vision, Neural Networks, Model Training, Model Evaluation
- **Frameworks & Libraries:** PyTorch, TensorFlow, Scikit-learn, OpenCV, MediaPipe, NumPy
- **Backend & APIs:** REST API Development, API Integration, JSON, Python-based API Development
- **Databases:** MySQL, SQLite, PostgreSQL
- **Software Engineering:** Object-Oriented Programming (OOP), Data Structures & Algorithms, Debugging, Software Testing, Git, GitHub
- **Development Tools & Platforms:** Linux, Jupyter Notebook, VS Code, IntelliJ IDEA, Flutter, BLoC
- **Additional:** Competitive Programming, Problem Solving, AI Application Development

EDUCATION

B.Sc. in Computer Science and Engineering (CSE)

Expected August-2026

- Shyamoli Engineering College (Constituent of University of Dhaka)

ACTIVITIES & ACHIEVEMENTS

- **Competitive Programming:** Active participant on Codeforces with a maximum rating of **1080**; regularly solve algorithmic problems involving data structures and algorithms.
- **ICPC 2023:** Participated in the International Collegiate Programming Contest (ICPC), solving complex algorithmic challenges under strict time constraints.